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Outdoor cooking station with warming chamber and method thereof

Abstract

An outdoor cooking station configured to keep cooked food warm within a warming chamber of the cooking station. The cooking station includes a main body with frame components and panels, the panels coupled to the frame components so as to extend and define the warming chamber, the warming chamber positioned below heating elements supported by the main body. The warming chamber is fixedly positioned relative to the heating elements. The main body includes a door pivotably coupled to the main body and positioned along a front side of the main body, the door pivotably moveable between an open position and a closed position. With this arrangement, the door facilitates access to the warming chamber to place cooked food therein.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/123,766, filed Mar. 20, 2023, which is a divisional of U.S. patent application Ser. No. 16/405,899, filed May 7, 2019, which claims the benefit of U.S. Provisional Application No. 62/669,500, filed May 10, 2018, the disclosures of which are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to outdoor type cooking stations and, more specifically, the present invention relates to maintaining cooked food in a warm state with a warming chamber integrated with an outdoor type cooking station.

BACKGROUND

(2) Barbequing has become a popular and pervasive tradition in much of the world. A barbeque grill is a device for cooking food by applying heat directly below a grill. There are several varieties of grills but most fall into one of two categories, either gas fueled or charcoal. Gas fueled grills typically use propane or natural gas as a fuel source, with the gas flame either cooking the food directly or heating grilling elements which in turn radiate the heat necessary to cook the food. Grilling has become a popular method of cooking food due to the unique flavors and texture imparted to the food during the grilling process.

(3) A griddle is a cooking device consisting of a broad flat surface that can be heated using a variety of means and is used in both residential and commercial applications for a variety of cooking operations. The griddle is most commonly a flat metal plate composed of cast or wrought iron, aluminum or carbon steel. Griddles are commonly heated directly or indirectly by open flame or electrical elements. Using a griddle placed directly on a barbeque grill or over flame burners has also become popular when cooking foods not as well suited for cooking directly on a grill over an open flame.

(4) Often large quantities of food are prepared outdoors with a griddle or grill type barbequing station. At such times, difficulties arise with keeping the already cooked food in a warm state. Barbequing stations with pivotably attached hoods often include one or more racks that, when the hood is opened, the racks are positioned above and rearward of the cooking surface and, when the hood is closed, the racks typically move above the cooking surface and within the hood. As food is being cooked, one will typically place the already cooked food on the racks while continuing to cook on the cooking surface. However, with the hood open, the rack's position relative to the cooking surface typically will not keep the food sufficiently warm. Rather, to keep the food warm, one must close the hood of the barbequing station, which ends up interrupting the cooking process. Further, when the hood is closed, the temperature quickly rises to undesirable temperatures for the food being cooked on the grill or griddle as well as the already cooked food on the rack, often resulting in burning the food on the rack and sometimes causing the food to become dry. As such, it would be advantageous to have the ability to keep already cooked food warm without interrupting the cooking process and without causing the cooking temperature of the cooking station to reach undesirable cooking temperatures.

BRIEF SUMMARY OF THE INVENTION

(5) The present invention is directed to various embodiments of an outdoor cooking station for cooking food and maintaining cooked food in a warmed state. In one embodiment, the outdoor cooking station includes a main body with frame components and panels, the panels coupled to the frame components so as to extend and define a warming chamber. The warming chamber is positioned directly below heating elements supported by the main body, the heating elements positioned below a cooking surface configured to be associated with the main body. Further, the warming chamber is fixedly positioned relative to the heating elements. The main body includes a door pivotably coupled to the main body and is positioned along a front side of the main body. With this arrangement, the door is pivotably moveable between an open position and a closed position such that, in the open position, the door exposes an opening defined in the front side of the main body to facilitate access to the warming chamber to place the cooked food therein.

(6) In another embodiment, the main body includes a baffle extending to define an upper side panel and a front facing vent, the upper side panel positioned to at least partially define an upper side of the warming chamber, the front facing vent having openings defined therein and positioned above the door. In another embodiment, the main body includes a baffle extending with an upper side panel and a front facing vent, the front facing vent extending perpendicular relative to the upper side panel, the front facing vent positioned above the door with openings defined therein and the upper side panel partially defining the warming chamber.

(7) In another embodiment, the door pivots adjacent a bottom edge of the opening defined in the front side of the main body. In another embodiment, the door is coupled to a hinge mechanism with a spring coupled thereto. In still another embodiment, the warming chamber includes a removable rack, the removable rack suspended over oppositely positioned ledges within the warming chamber.

(8) In another embodiment, the warming chamber is at least partially defined by a bottom panel, a first panel, a second panel, and a rear panel, the first panel being positioned opposite the second panel, the rear panel being positioned opposite the door in the closed position, the warming chamber being at least partially open and exposed to the heating elements positioned above the warming chamber. In a further embodiment, the warming chamber includes one or more inside panels extending laterally relative to the rear panel, the one or more inside panels defining one or more vents therein to facilitate convection heat and radiation heat from the heating elements to heat the warming chamber. In another further embodiment, the warming chamber includes one or more inside panels extending laterally relative to the rear panel, the one or more inside panels extending to define an opening to facilitate convection and radiation heat from the heating elements to heat the warming chamber.

(9) In another embodiment, the cooking surface includes a planar griddle surface. In yet another embodiment, the main body includes at least two legs with caster wheels coupled thereto.

(10) In accordance with another embodiment of the present invention, an outdoor cooking station is provided. The outdoor cooking station includes a main body with frame components and panels, the panels coupled to the frame components so as to extend and define an upper chamber region and a lower chamber region. The upper chamber region includes multiple heating elements supported by the main body and disposed below an upper edge of the main body. The upper edge extends adjacent a cooking surface configured to be associated with the main body. The lower chamber region is fixedly positioned relative to the upper chamber region and disposed directly below the upper chamber region and the heating elements. The panels of the lower chamber region extend to define a bottom panel, a rear panel, a first panel and a second panel, the first panel being positioned opposite the second panel, the rear panel being positioned opposite a door pivotably coupled to the main body. The door is pivotably moveable to an open position to facilitate access to the lower chamber region.

(11) In another embodiment, the main body includes a front facing vent panel with openings defined therein, the front facing vent panel positioned directly above the door. In another embodiment, the main body includes a baffle extending with an upper side panel and a front facing

vent, the front facing vent extending perpendicular relative to the upper side panel, the front facing vent positioned above the door with openings defined therein and the upper side panel at least partially defining the warming chamber.

(12) In another embodiment, the door is coupled to a hinge mechanism with a spring coupled thereto such that the door is biased to a closed position. In another embodiment, the lower chamber region includes a rack suspendably positioned between the first and second panels.

(13) In another embodiment, the upper chamber region is separated from the lower chamber region with one or more inside panels, the one or more inside panels defining one or more vents therein to facilitate convection and radiation heat from the heating elements to heat the lower chamber region. In still another embodiment, the upper chamber region is separated from the lower chamber region with one or more inside panels, the one or more inside panels extending to define an opening along an edge of the one or more inside panels, the opening configured to supply heat to the lower chamber region from the heating elements. In another embodiment, the cooking surface includes a planar griddle surface.

(14) In accordance with another embodiment of the present invention, a method for warming cooked food is provided. The method includes the steps of: providing an outdoor cooking station with a main body with frame components and panels, the panels coupled to the frame components so as to extend and define a warming chamber, the warming chamber positioned below heating elements supported by the main body, the heating elements positioned below a cooking surface configured to be associated with the main body, the warming chamber fixedly positioned relative to the heating elements, the main body including a door pivotably coupled to the main body and positioned along a front side of the main body, the door pivotably moveable between an open position and a closed position such that, in the open position, the door exposes an opening defined in the front side of the main body to facilitate access to the warming chamber to place the cooked food therein; heating the cooking surface with the heating elements for cooking food on the cooking surface; pivotably opening the door so as to expose an opening defined in the front side of the main body to facilitate access to the warming chamber; placing cooked food through the opening and within the warming chamber; and pivotably closing the door to the closed position to keep the cooked food warm within the warming chamber.

(15) In another embodiment, the method further includes the step of facilitating air flow between the front side of the main body and the heating elements through a front facing vent positioned above the door. In another embodiment, the providing step includes providing a baffle to at least partially define an upper side of the warming chamber such that the baffle extends with a front facing vent positioned directly above the door.

(16) In another embodiment, the method step of providing includes providing the door with a hinge and spring arrangement so that the door is biased toward the closed position. In another embodiment, the method step of providing includes providing the warming chamber with a removable rack positioned between the first and second panels. In another embodiment, the method step of heating includes supplying heat to the warming chamber with convection and radiation heat from the heating elements. In still another embodiment, the providing step includes providing one or more inside panels extending to define a separation between the heating elements and the warming chamber such that the one or more inside panels define one or more openings to facilitate supplying heat from the heating elements to the warming chamber.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The foregoing and other advantages of the invention will become apparent upon reading the following detailed description and upon reference to the drawings in which:

- (2) FIG. 1 is a right perspective view of an outdoor cooking station with a griddle member positionable over the cooking station, depicting a door in an open position to expose a warming chamber integrated within the cooking station, according to an embodiment of the present invention;
- (3) FIG. 2 is a left perspective view of the outdoor cooking station, depicting the door in a closed position, according to another embodiment of the present invention;
- (4) FIG. 3 is a top perspective view of the outdoor cooking station without the griddle member, depicting heating elements positioned within a main body of the outdoor cooking station, according to another embodiment of the present invention;
- (5) FIG. 4 is a top view of the cooking station, according to another embodiment of the present invention;
- (6) FIG. 5 is a cross-sectional view taken along section A-A of FIG. 4 of the main body of the outdoor cooking station, depicting an upper chamber region above a lower chamber region of a main chamber, according to another embodiment of the present invention;
- (7) FIG. 6 is a perspective bottom view of the outdoor cooking station with the door in the open position, according to another embodiment of the present invention;
- (8) FIG. 7 is a rear view of the outdoor cooking station, according to another embodiment of the present invention;
- (9) FIG. 8 is a front view of the outdoor cooking station with the door in the open position, according to another embodiment of the present invention;
- (10) FIG. 9 is an enlarged partial view of the door, according to another embodiment of the present invention;
- (11) FIG. 10 is a perspective view of another embodiment of an outdoor cooking station with a griddle member positioned thereon, depicting the outdoor cooking station having a warming chamber with a door in the closed position, according to the present invention;
- (12) FIG. 11 is a perspective view of the outdoor cooking station of FIG. 10, depicting the door of the warming chamber in the open position, according to another embodiment of the present invention;
- (13) FIG. 12 is a perspective bottom view of the outdoor cooking station of FIG. 10, depicting a baffle positioned within the main body of the cooking station, according to another embodiment of the present invention; and
- (14) FIG. 13 is a cross-sectional view taken along section line A-A of FIG. 10, according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(15) Referring to FIGS. 1 and 2, an outdoor cooking station **10** with a warming chamber **12**, according to one embodiment of the present invention, is provided. The outdoor cooking station **10** may be sized and configured to be employed with a griddle member **14** and/or a grill type cooking surface or any other suitable cooking member or cooking surface for cooking food on the outdoor cooking station **10**. The warming chamber **12** may be integrated with a region or chamber holding heating elements **16** (FIG. 3) of the cooking station **10** for cooking food. Further, the warming chamber **12** may be accessed with a door **18** that may be moved between an open position (FIG. 1) and a closed position (FIG. 2). In this manner, upon cooking food with the outdoor cooking station **10**, the door **18** may be opened for placing cooked food within the warming chamber **12** and then moved to the closed position to maintain the heat within the warming chamber **12** and within the cooked food.

(16) The outdoor cooking station **10** may include a main body **20** with various frame components **22** and panels **24** coupled to the frame components **22**. The main body **20** may extend to define a front side **26**, a rear side **28**, a top side **30**, a bottom side **32**, a first side **34** and a second side **36**. The first side **34** may be oppositely positioned relative to the second side **36** and may also be referenced as respective left and right sides. Similarly, the front side **26** may be oppositely

positioned relative to the rear side **28**, and the bottom side **32** may be oppositely positioned relative to the top side **34**. Each of the front side **26**, rear side **28**, bottom side **32**, and first and second sides **34**, **36** may include one or more of the panels **24** and the frame components **22** that extend to surround and define the warming chamber **12** and heating elements **16** of the cooking station **10**. The various frame components **22** may be coupled together to extend horizontally and vertically, as known by one in the art, to form the main body **20** with the panels **24** secured to the frame components **22**, as previously set forth. Similarly, the panels **24** may extend horizontally between first and second sides **34**, **36** and front and rear sides **26**, **28** of the main body **20** as well as extend vertically between a lower end **38** or lower portion and an upper edge **40** of the main body **20**.

(17) In another embodiment, the frame components **22** may extend downward from the main body **20** to exhibit legs **42** of the main body **20** with various cross-members **44** between the legs **42**. Such various cross-members **44** may extend and be secured along the lower end **38** of the main body **20** and, for example, extend adjacent ends **46** of the legs **42** to stabilize the structure of the outdoor cooking station **10**. Further, the cooking station **10** may include caster wheels **48** securable to the ends **46** of the legs **42** to facilitate portability of the outdoor cooking station **10**. As such, the outdoor cooking station **10** may be readily moveable with minimal effort so that the cooking station **10** is portable. Other components of the cooking station **10** may include one or more handles **50** coupled to one of the first and second sides **34**, **36** of the main body **20**. Further, the cooking station **10** may include a first side shelf **52** and a second side shelf **54** coupled to the first side **34** and second side **36** of the main body **20**, respectively. In addition, the cooking station **10** may include a propane tank holder **56** coupled to one or more of the cross-members **44**, the propane tank holder **56** configured to mount a propane tank (not shown) thereto.

(18) With reference to FIGS. **1**, **3** and **4**, the top side **30** of the main body **20** may include the upper edge **40** extending with some of the frame components **22** in a rectangular configuration so as to exhibit a rectangular periphery. The rectangular periphery of the upper edge **40** may also define a square periphery. The upper edge **40** may be disposed adjacent a cooking surface **60**, such as the flat planar surface of the griddle member **14** associated therewith or any other type of cooking surface **60**, such as a grill type cooking surface. The upper edge **40** may be sized and configured to receive, for example, small legs **62** of a griddle member **14** so that the upper edge **40** may support and stabilize the griddle member **14** thereon. In this manner, the upper edge **40** of the main body **20** and/or the cooking station **10** may be sized and configured to receive the griddle member **14**. Further, in another embodiment, the cooking station **10** may include a hood (not shown) associated therewith that may be pivotably coupled to the rear side **28** of the main body **20** to be closeable over the upper edge **40** and over the cooking surface **60**. Such hood may also be removable from the cooking station **10** without being pivotably coupled thereto.

(19) As previously set forth, the front side **26** of the main body **20** may include the door **18** for accessing the warming chamber **12**. Further, the front side **26** of the main body **20** may include multiple burner control valves **64** or knobs positioned over a front panel **66**. Each of the burner control valves **64** or knobs may be associated with a heating element **16** or flame burners. The heating elements **16** may extend between the front side **26** and rear side **28** of the main body **20** below the upper edge **40** of the main body **20** such that the heating elements **16** may be secured to frame components **22** extending within and adjacently along the front and rear sides **26**, **28** of the main body **20**. Each of the heating elements **16** may hold structure for feeding pressurized fuel therethrough, such as propane gas or natural gas, and controlled with burner control valves **64** or knobs. In addition, the front panel **66** may include an ignitor switch **68** that may be depressed to provide a spark to ignite the gas fueled heating elements **16**. The cooking station **10** may include typical components to facilitate igniting the heating elements **16** and heating the cooking surface **60** (FIG. **1**) associated with the cooking station **10**, such as various valves, tubing, manifold, gas couplings, fasteners, and any other components, as known to one of ordinary skill in the art.

(20) Now with reference to FIGS. **5**, **6** and **7**, the warming chamber **12** of the outdoor cooking

station **10** will now be described. As previously set forth, the main body **20** of the outdoor cooking station **10** may include the panels **24** extending over the frame components **22** to define the warming chamber **12**. Such panels **24** may be coupled to the frame components **22** to define a main chamber **70** extending to define a lower chamber region **72** and an upper chamber region **74**, the lower chamber region **72** being directly below the upper chamber region **74**, each chamber generally shown in dashed lines in FIG. 5. The upper chamber region **74** may include the heating elements **16** extending across the upper chamber region **74**, such as between the front and rear sides **26**, **28** of the main body **20**. The lower chamber region **72** and/or the main chamber **70** may be referenced as the before described warming chamber **12**. Further, the lower chamber region **72** or warming chamber **12** may be fixed relative to the heating elements **16** such that the lower chamber region **72** or warming chamber **12** does not move or pivot or is not associated with a drawer or the like. The main chamber **70** may be defined with the front panel **66** or other front side extending panels, a rear panel **76**, a bottom panel **78**, and first and second side panels **80**, **82**. Each of the front panel **66**, the rear panel **76**, the bottom panel **78**, and the first and second side panels **80**, **82** may each extend with one or more panels defining such sides. For example, the one or more front side **26** or front panels **66** may extend with one front panel having the burner control valves **64** therewith and another front panel may include a door opening **84** defined therein. Such door opening **84** defined in one of the front panels **66** may include the door **18** pivotably coupled to the main body **20** that may be moveable between the open and closed positions, as previously set forth. Further, such door opening **84** may be sized and configured for a user to readily place cooked food on a tray or the like within the warming chamber **12**. Further, some of the first and second side panels **80**, **82** and some of the rear panels **76** may include vent structure **86** to prevent the warming chamber **12** from becoming too hot or over-heating, for example.

(21) With reference to FIGS. 4 and 5, the upper chamber region **74** may be separated by one or more inside panels, such as a front inside panel **88** and a rear inside panel **90** each extending between the first and second side panels **80**, **82**. Such one or more inside panels may include one or more openings, such as vents **92**, defined in the one or more inside panels. The front and rear inside panels **88**, **90** may each define an inner periphery **94** or edge to define an opening **96** between the respective inner peripheries **94** of the front and rear inside panels **88**, **90**. In this manner, the vents **92** defined in the one or more inside panels and the opening **96** may be employed to assist heating the lower chamber region **72** with convection and radiation heat supplied by and emanating from the heating elements **16**. Further, the front and rear inside panels **88**, **90** may be heated by the heating elements **16** to then provide radiation heat to the lower chamber region **72**.

(22) Now with reference to FIGS. 1, 8 and 9, the lower chamber region **72** or warming chamber **12** of the cooking station **10** may include a removable rack **100** or grate like structure. Such removable rack **100** may be suspended over first and second brackets **102**, **104** secured to inside surfaces of the main body **20**, each bracket **102**, **104** including a ledge for the rack **100** to be disposed or rest upon. The rack **100** may be sized to extend a majority of the expanse within the lower chamber region **72** between the first and second side panels **80**, **82** and between the rear panel **76** and the door **18**. The rack **100** may extend horizontally level and parallel with the bottom panel **78** within the warming chamber **12**. Further, the rack **100** may be suspended so as to be positioned a height **105** above the bottom panel **78**. In another embodiment, additional brackets may be secured to inside surfaces of the main body **20** so that the rack **100** may be positioned at various heights above the bottom panel **78**. In one embodiment, the height **105** of the rack **100** may be positioned to be about half the height of the door opening **84**. Such rack **100** may be sized and configured for placing, for example, a tray or the like of cooked food on the rack **100** for maintaining the food in a warm state.

(23) Further, as previously set forth, the door **18** may be pivotably moveable between the open and closed positions. The door **18** may include an inner surface **106** and an outer surface **108** each extending to a rectangular periphery. The outer surface **108** of the door **18** may include a door

handle **110** positioned thereon to assist the user to manually move the door **18** to the open and closed positions. The door **18** may be pivotably coupled to the main body **20** with a hinge mechanism **112**. For example, the hinge mechanism **112** may include a first elongated part **114** and a second elongated part **116**, the first and second elongated parts **114**, **116** rotatably coupled together at end portions thereof. At the other end of the first elongated part **114**, the first elongated part **114** may be rotatably coupled to a first hinge bracket **118** secured to the main body **20**. At the other end of the second elongated part **116**, the second elongated part **116** may be rotatably coupled to a second hinge bracket **120** along a mid end portion of the inner surface **106** of the door **18**. In one embodiment, such hinge mechanism **112** may include two hinge mechanisms, one coupled adjacent to each of the opposing inner side portions of the main body **20** and door **18**. In this manner, the door **18** may be pivotably coupled to the main body **20** so as to pivot about a bottom edge **122** of the door **18**. Further, each of the two hinge mechanisms **112** may include a spring **124** associated therewith. Such spring **124** may be coupled between a mid portion of the hinge mechanism **112** and the main body **20** and, more specifically, the spring **124** may extend between an aperture defined in the second elongated part **116** of the hinge mechanism **112** and a bracket aperture defined in the first hinge bracket **118**. With this arrangement, the door **18** may be spring biased toward the closed position.

(24) Further, the inner surface **106** of the door **18** may include a protrusion **126** sized and configured to cooperate with a roller catch **128** mounted to the main body **20** within the lower chamber region **72** (see FIG. 5). Such protrusion **126** may define a diamond shaped profile and may be positioned adjacent an upper corner portion of the inner surface **106** of the door **18**. The protrusion **126** may cooperate with the roller catch **128** such that the roller catch **128** may include two inward spring biased rollers. With this arrangement, upon closing the door **18**, the diamond shaped protrusion **126** is pushed against the roller catch **128** between the rollers such that the protrusion **126** separates the rollers and is caught between the rollers in a spring biased manner. In one embodiment, the inner surface **106** of the door **18** may include two protrusions **126**, one at opposite upper corner portions of the door **18** that correspond with two roller catches **128** at opposite upper side and forward positions of the warming chamber **12**. In this manner, the door **18** may be maintained in the closed position with both the spring **124** biasing the door **18** in the closed position as well as being held shut with the protrusion **126** and roller catch **128** arrangement.

(25) Now with reference to FIGS. 1 and 2, a user may ignite the heating elements **16** (FIG. 3) with the igniter switch **68** after opening the valve of, for example, a propane tank and turning one or more of the burner control valves **64** to an on position. As the user is heating the cooking surface **60** of, for example, a griddle member **14**, the heating elements **16** also supply convection heat and radiation heat to the warming chamber **12** below the heating elements **16**. Further, the heating elements **16** may provide heat to the front and rear inside panels **88**, **90**, which may then provide radiation heat to the lower chamber region **72** or warming chamber **12**. As the user completes cooking a batch of food, the user may open the door **18** of the warming chamber **12** and place the cooked food over the rack **100** in the warming chamber **12**. The user may then close the door **18** of the warming chamber **12** to maintain and build the heat within the warming chamber **12**. The user may then continue to cook food on the griddle member **14** while also maintaining the already cooked food in a warm state. In this manner, the warming chamber **12** of the outdoor cooking station **10** may be employed to maintain cooked food in a warm state while continuing to actively cook food on the griddle member **14**.

(26) Now with reference to FIGS. 10-13, another embodiment of an outdoor cooking station **130** with a warming chamber **132** is provided. The outdoor cooking station **130** may include similar structural features as described in the previous embodiment including a main body **134** defined with frame components **136** and panels **136** sized to support heating elements (not shown) and a griddle member **140**. Further, the outdoor cooking station **130** may include a door **142** for accessing the warming chamber **132**. In this embodiment, the warming chamber **132** may at least partially be

defined with a baffle **144**. As in the previous embodiment, the warming chamber **132** may be defined with a bottom panel **146**, a rear panel **148**, and first and second side panels **150**, **152**, each of which may define an external portion of the main body **134**. Further, the warming chamber **132** may also be defined with the door **142** that may be hingably opened and closed (see FIGS. **10** and **11**) along a bottom side **154** of the door **142** to expose a chamber opening **160** and the warming chamber **132**, the chamber opening **160** may be defined by front side edges of each of the baffle, bottom panel **146**, and first and second side panels **150**, **152**. Along an upper side of the warming chamber **132**, such upper side may be exposed to the heating elements, similar to that described in the previous embodiment, but also may include the baffle **144** along the upper side to at least partially shield the warming chamber **132** from some gas carrying components of the cooking station **130** as well as direct heat to the warming chamber.

(27) With reference to FIGS. **11-13**, the baffle **144** may include an upper side panel **162** and a front facing vent **164**. In one embodiment, the upper side panel **162** may be integrally formed with the front facing vent **164**. The upper side panel **162** may be sized and configured to extend and at least partially define the upper side or ceiling of the warming chamber **132**. The baffle **144** or upper side panel **162** may extend with an elongated length, a width and a thickness, the thickness being a thickness of, for example, typical sheet metal. The baffle **132** may extend with a front edge **166**, a rear edge **168** and first and second edges **170**, **172**, the first edge **170** being opposite the second edge **172** and the front edge **166** being opposite the rear edge **168**. The baffle **144** or upper side panel **162**, adjacently along the first and second edges **170**, **172** of the upper side panel **162**, may be supported by supporting members **174** fixated to an internal surface, such as the first and second side panels **150**, **152**, of the warming chamber **132**. With the upper side panel **162** positioned over the supporting members **174**, the upper side panel **162** may extend rearward toward rear panel **148**, but only extend to about a mid-point or intermediate point rearward along the main body **134** so as to leave a gap **176** between the rear edge **168** and the rear panel **148**. As such, the upper side panel **162** extends with the width so as to leave the gap **176** so that the warming chamber **132** may be directly exposed to the heating elements positioned above the baffle **144** and within the main body **134**. The rear edge **168** of the baffle **144** may also include an upstanding portion **177** extending upward along the length of the rear edge **168** of the upper side panel **162**. Such upstanding portion **177** may extend upward to a height similar to the front facing vent **164** or shorter. Further, the upstanding portion **177** may extend parallel relative to the front facing vent **164** and may be oppositely positioned relative to the front facing vent such that the front edge **166** of the upper side panel **162** exhibits the front facing vent **164** and the rear edge **168** exhibits the upstanding portion **177**. In this manner, the upper side panel **162** and the upstanding portion **177** of the baffle **144** may be sized and configured to direct heat to the warming chamber **132** and at least partially shield and protect the gas carrying components of the outdoor cooking station **130** from heat so as to maintain a safe temperature.

(28) The front facing vent **164** may be sized and configured to be positioned along the front edge **166** of the upper side panel **162**. The front facing vent **164** may be an elongated flat structure extending horizontally directly above the front edge **166** of the upper side panel **162** and may extend upward toward a front panel **178**, the front panel **178** holding control valves **179** of the outdoor cooking station **130**. Further, the front facing vent **164** may be inset inward relative to the front panel **178**. The front facing vent **164** may define multiple openings **180** therein, the openings **180** extending through the front facing vent **164** and sized and configured to vent external air therethrough. The multiple openings **180** may be positioned in an aligned arrangement so as to extend in a single row horizontally along the front facing vent **164**. The multiple openings **180** may each exhibit a circular profile, such as an oval type profile, that may be elongated horizontally. In another embodiment, the multiple openings **180** of the front facing vent **164** may be positioned with multiple elongated slots defined in the front facing vent, or with any other suitable profile that facilitates air flow therethrough.

(29) The front facing vent **164** may be integrally formed with the upper side panel **162** so as to define a bend **182** between the upper side panel **162** and the front facing vent **164**. The front facing vent **164** may extend with a planar structure that may extend substantially orthogonal relative to the upper side panel **162** of the baffle **144**. In another embodiment, the front facing vent **164** may be positioned substantially parallel relative to the front panel **178** of the main body **134** so as to be offset relative to the front panel **178**. In another embodiment, the front facing vent **164** may be positioned along the front side of the main body **134** with the upper side panel **162** extending therefrom at an acute or obtuse angle, instead of the orthogonal arrangement, previously set forth. In another embodiment, the front facing vent **164** may be fixed to the upper side panel **162** with fasteners. In another embodiment, the front facing vent **164** may be fixated to the main body **134** with the fasteners such that a lower edge of the front facing vent **164** extends alongside the front edge **166** of the upper side panel **162**. With this arrangement, the front facing vent **164** and the upper side panel **162** may be employed to protect various components of the outdoor cooking station **130** from over-heating. Further, the multiple openings **180** defined in the front facing vent **164** may be sized and configured facilitate external air-flow to the gas carrying components of the cooking station, such as the tubing **181**, manifold, valve components, etc., to maintain a safe temperature for such components. For example, the tubing **181** may extend in a space **185** defined above the upper side panel **162** and between the front facing vent **164** (adjacent the openings **180**) and the upstanding portion **177** of the baffle **144**. As such, the baffle **144** may direct heat to the warming chamber **182** as well as define the space **185** for maintaining a safe temperature for gas carrying components of the outdoor cooking station **130**.

(30) Further, the first and second side panels **150**, **152** and the rear panel **148** of the warming chamber **132** may also exhibit vents **183** formed therein to ensure that an appropriate amount of external air flow may fuel the gas flame of the heating elements. This is especially important for outdoor cooking stations having a griddle member positioned thereon or for when a hood is covering a top side of a given cooking station.

(31) As depicted in FIG. **11**, the warming chamber **132** may not include a removable shelf therein, as described in previous embodiments. Rather, the bottom panel **146** of the warming chamber **132** may facilitate supporting food or the like that the user may desire to keep warm. Further, with reference to FIGS. **10** and **11**, the main body **134** may include first and second side shelves **184**, **186** that may be substantially level with an upper edge **188** of the main body **134**. Each of the first and second side shelves **184**, **186** may include an elongated opening **190** and ridge **192** arrangement to discourage users of the outdoor cooking station **130** from placing objects on an upper surface of the side shelves that may be too close to the griddle member **140** and the hot air flowing out from under the griddle member **140**, similar to that disclosed in commonly owned U.S. patent application Ser. No. 16/364,012, filed on Mar. 25, 2019, the entire contents of which are incorporated herein by reference. Further, the first and second side shelves **184**, **186** may include various hooks **194** and/or a magnetic strip **196** along a readily accessible surface (other than an upward facing surface) of the first and second side shelves **184**, **186** for supporting, for example, cooking utensils. Further, the door **142** may be biased to a closed position with a spring (not shown) such as that described in the previous embodiment, or the door **142** may be maintained in the closed position with magnets positioned or integrated within a top edge of the door and/or adjacent the baffle or with any other suitable coupling structure or means for holding the door **142** in the closed position.

(32) The various structural components of the outdoor cooking station described herein and any other structural components thereof or associated therewith may be formed of various metallic materials, such as steel, stainless steel, copper, aluminum or any other suitable material with high temperature ratings, such as various suitable polymeric materials, and may be formed from known structural components, such as sheet metal at various gauges/thicknesses or other known metallic structures, such as tubing or the like, and may be formed and manufactured through various known

processes and techniques known in the art, such as casting, welding, rolling, bending, pressing, fastening, etc., as known by one of ordinary skill in the art.

(33) While the invention may be susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. Further, the structural features of any one embodiment disclosed herein may be combined or replaced by any one of the structural features of another embodiment set forth herein. However, it should be understood that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention includes all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the following appended claims.

Claims

1. An outdoor cooking station for cooking food and maintaining cooked food in a warmed state, the outdoor cooking station comprising: a main body with frame components and panels extending to define a front side, a rear side, a left side, and a right side, the panels coupled to the frame components so as to extend and define a warming chamber, the warming chamber positioned below gas flame burners supported by the main body and extending between the front side and the rear side, the gas flame burners positioned below a griddle having a flat cooking surface configured to be supported by the main body, the warming chamber positioned below the gas flame burners, the main body including a door movable relative to the main body and positioned along the front side of the main body, the door moveable between an open position and a closed position such that, in the open position, the door exposes an opening defined in the front side of the main body to facilitate access to the warming chamber to place the cooked food therein; wherein the main body comprises a baffle extending to define an upper side panel and a front facing vent, the upper side panel positioned to at least partially define an upper side of the warming chamber, the front facing vent having openings defined therein and positioned above the door so that the front facing vent extends upward from the upper side panel.
2. The outdoor cooking station of claim 1, wherein the front facing vent extending substantially perpendicular relative to the upper side panel.
3. The outdoor cooking station of claim 1, wherein the door is moveable to pivot adjacent a bottom edge of the opening defined in the front side of the main body.
4. The outdoor cooking station of claim 1, wherein the warming chamber comprises a removable rack, the removable rack suspended over oppositely positioned ledges within the warming chamber.
5. The outdoor cooking station of claim 1, wherein the warming chamber is at least partially defined by a bottom panel, a first panel, a second panel, and a rear panel, the first panel being positioned opposite the second panel, the rear panel being positioned opposite the door in the closed position, the warming chamber being at least partially open and exposed to the gas flame burners positioned above the warming chamber.
6. The outdoor cooking station of claim 5, wherein the warming chamber comprises one or more inside panels extending laterally relative to the rear panel, the one or more inside panels extending to define an opening to facilitate convection and radiation heat from the gas flame burners to heat the warming chamber.
7. An outdoor cooking station, comprising: a main body with frame components and panels extending to define a front side, a rear side, a left side, and a right side, the panels coupled to the frame components so as to extend and define an upper chamber region and a lower chamber region, the upper chamber region including multiple gas flame burners supported by the main body extending between the front side and the rear side and disposed below an upper edge of the main body, the upper edge extending adjacent a griddle having a flat cooking surface configured to be supported by the main body, the lower chamber region fixedly positioned relative to the upper

chamber region and disposed directly below the upper chamber region and the gas flame burners, the panels of the lower chamber region extending to define a bottom panel, a rear panel, a first panel and a second panel, the first panel being positioned opposite the second panel, the rear panel being positioned opposite a door movable relative to the main body, the door moveable to an open position to facilitate access to the lower chamber region; wherein the main body comprises a baffle extending to define an upper side panel and a front facing vent, the upper side panel positioned to at least partially define an upper side of the warming chamber, the front facing vent having openings defined therein and positioned above the door so that the front facing vent extends upward from the upper side panel.

8. The outdoor cooking station of claim 7, wherein the door is coupled to a hinge mechanism with a spring coupled thereto such that the door is biased to a closed position.

9. The outdoor cooking station of claim 7, wherein the lower chamber region comprises a rack suspendably positioned between the first and second panels.

10. The outdoor cooking station of claim 7, wherein the upper chamber region is separated from the lower chamber region with one or more inside panels, the one or more inside panels defining one or more vents therein to facilitate convection and radiation heat from the gas flame burners to heat the lower chamber region.

11. The outdoor cooking station of claim 7, wherein the upper chamber region is separated from the lower chamber region with one or more inside panels, the one or more inside panels extending to define an opening along an edge of the one or more inside panels, the opening configured to supply heat to the lower chamber region from the gas flame burners.

12. The outdoor cooking station of claim 7, wherein the door is moveable to pivot relative to the main body such that, upon the door being moved to the open position, the door exposes an opening defined in the front side of the main body, the door being pivotable adjacent a bottom edge of the opening defined in the front side of the main body.
